

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built a multi-agent production system via OpenAI Agents SDK and MCP tool-use, integrating services into large-scale workflows handling millions of daily transactions at 99%+ uptime.
- Engineered a pluggable distributed executor routing pipeline nodes across Docker and AWS Lambda, backed by an S3 artifact store eliminating redundant recomputation.
- Designed stateful recovery via Redis checkpointing so interrupted workflows resume deterministically from the last committed step across distributed execution boundaries.
- Owned full-stack delivery of backend services and APIs (Python/FastAPI, React/TypeScript); replaced polling with SSE for live metrics streaming, cutting redundant API calls 90%.
- Deployed a production service on AWS SageMaker for real-time decision support at sub-200ms p99 latency with RTO under 2 minutes.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut p99 service latency 420ms to 140ms across 50K+ daily requests by migrating synchronous serving to async vLLM on GCP Cloud Run with autoscaling.
- Built PySpark data processing jobs for 10K+ large-scale feature dataset preparation feeding downstream training and evaluation pipelines.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed systems for 1M+ users, addressing concurrency, consistency, and sharding, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS, DynamoDB).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.
- Standardized failure-handling contracts across 25+ financial services, reducing production escalations 60% in one quarter via consistent retry, timeout, and propagation semantics.
- Designed public-facing REST and gRPC APIs with CLI tooling, prioritizing backward compatibility and usability for partner integrations.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed KV store handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes root cause reports.

LangFetch

tinyurl.com/LangFetch

- Agent decomposes multi-table questions into dependency-ordered SQL subqueries with intermediate result passing; deployable as an MCP-compatible server for enterprise data queries.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, C++, TypeScript, JavaScript, SQL

Core CS: Data Structures and Algorithms, Operating Systems, Computer Networks

Distributed Systems & Data: Distributed Systems, REST/gRPC APIs, PostgreSQL, Redis, DynamoDB, BigQuery, NoSQL

Backend & Testing: FastAPI, Spring Boot, Microservices, JUnit, Mockito

Cloud & DevOps: AWS (EC2, S3, SageMaker, Lambda), GCP (Cloud Run, Vertex AI), Docker, Kubernetes, CI/CD

CERTIFICATIONS

Google Cloud Professional Machine Learning Engineer | AWS Certified Developer Associate